



COURSE TITLE (i.e., EE809 Analog IC Design)
MSEE (ICS) 2026

Tutorial # 2

DC I/V characteristics of MOS Transistors

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<u>Date</u>	<u>20th September, 2026</u>



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1. Overview

The tutorial itself walked through the DC characterization of an NMOS (nch_mac, W = 200 nm, L = 60 nm) in ADE Explorer. A dc sweep of V_{GS} at fixed V_{DS} gave the transfer curve and g_m was extracted from it with the Calculator deriv function; a parametric sweep of V_{GS} over a V_{DS} sweep gave the family of output curves; the saved operating-point code M0:region exposed the subthreshold window below V_{TH} ; and a body-bias sweep showed V_{TH} rising as the bulk was pulled below the source.

The exercise activities repeat that whole flow on a PMOS, and everything below was run on the pch_mac testbench rebuilt from the NMOS one. Activities 2.1 and 2.2 reproduce the transfer and output characteristics and ask why the current keeps rising after saturation, 2.3 locates the subthreshold region for the PMOS, and 2.4 measures how the threshold voltage moves with body bias and with temperature. Each section below states the analysis setup, shows the resulting plot and reads the physics off it.

2. Activities (DC I/V Characterization of a PMOS Transistor)

2.1. PMOS Testbench and sign convention

Source and bulk (n-well) of the pch_mac sit at $V_{DD} = 1.8$ V, and both characterization sources are referenced to V_{DD} rather than ground, so that:

$$V(\text{gate}) = V_{DD} - V_{SG} \quad V(\text{drain}) = V_{DD} - V_{SD}$$

The design variables V_{SG} and V_{SD} are therefore the source-to-gate and source-to-drain voltages directly. ADE plots the current into the drain terminal, which a PMOS pushes outward, so every trace is negative and magnitudes are quoted below.

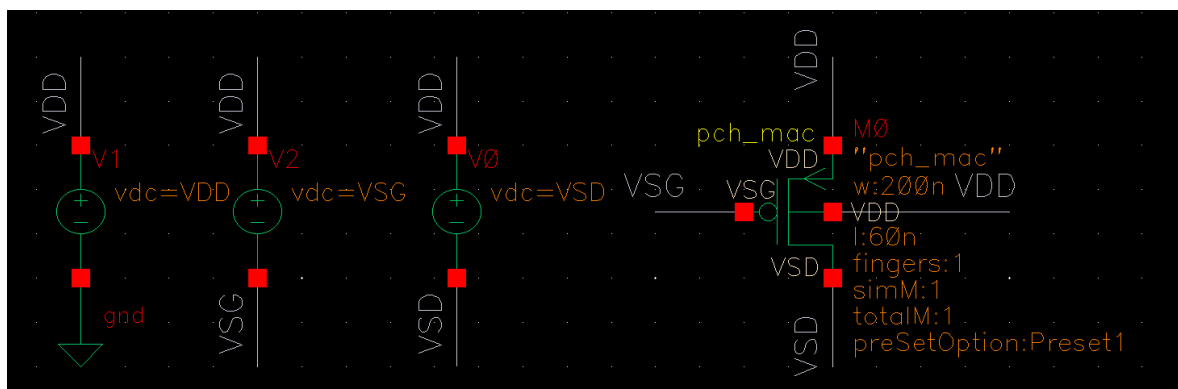


Figure 1: PMOS characterization testbench. V1 sets the rail, V2 and V0 sit between V_{DD} and the gate and drain nodes so that their values are V_{SG} and V_{SD} directly



Table 1 Operating point of M0 at $V_{SG} = 0.8$ V, $V_{SD} = 0.5$ V, $V_{DD} = 1.8$ V, 27 °C

Quantity	Symbol	Value	Quantity	Symbol
Threshold voltage	V_{TH}	-537 mV	Threshold voltage	V_{TH}
Saturation voltage	$ V_{SD,sat} $	209 mV	Saturation voltage	$ V_{SD,sat} $
Early voltage	V_A	1.28 V	Early voltage	V_A

Table 1 is the operating point from Results Browser, dcOp, at $V_{SG} = 0.8$ V and $V_{SD} = 0.5$ V.

2.2. Activity 2.1: I_D versus V_{SG} at constant V_{SD}

DC sweep of V_{SG} from 0 to 1 V with V_{SD} fixed at 0.5 V.

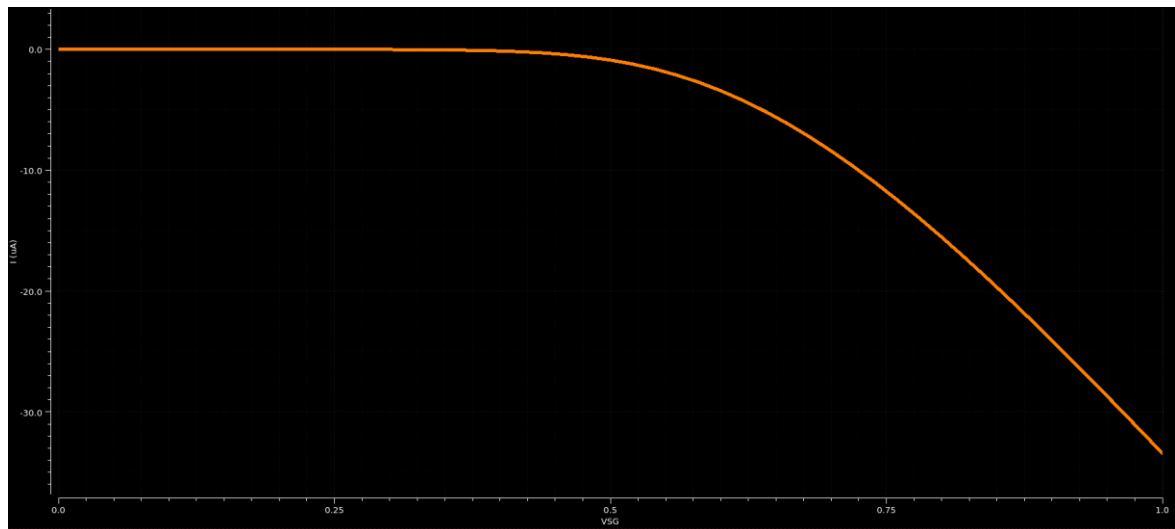


Figure 2. Transfer characteristic: drain current against V_{SG} at $V_{SD} = 0.5$ V.

Conduction starts at $|V_{TH}| = 537$ mV and reaches about 35 μ A at $V_{SG} = 1$ V. The overdrive never exceeds $V_{SD} = 0.5$ V, so the device stays in saturation over the conducting part of the sweep (region = 2 in Fig. 5) and the trace is:

$$I_D = \frac{1}{2} \mu_p C_{ox} (W/L) (V_{SG} - |V_{TH}|)^2$$

The rise is sub-quadratic: doubling the overdrive from about 0.21 V to 0.46 V raises the current roughly three times instead of the 4.7 the square law predicts, which is velocity saturation again.

The transconductance is the slope of this curve,

$$g_m = \partial I_D / \partial V_{SG} \quad \text{at constant } V_{SD}$$

taken with the Calculator deriv function or read as M0:gm: 81.1 μ S at the same bias, where $I_D = 15.53$ μ A, so $g_m/I_D = 5.2$ V⁻¹.



2.3. Activity 2.2: I_D versus V_{SD}

2.3.1 Single V_{SG}

V_{SD} swept from 0 to 1 V at $V_{SG} = 0.8$ V.

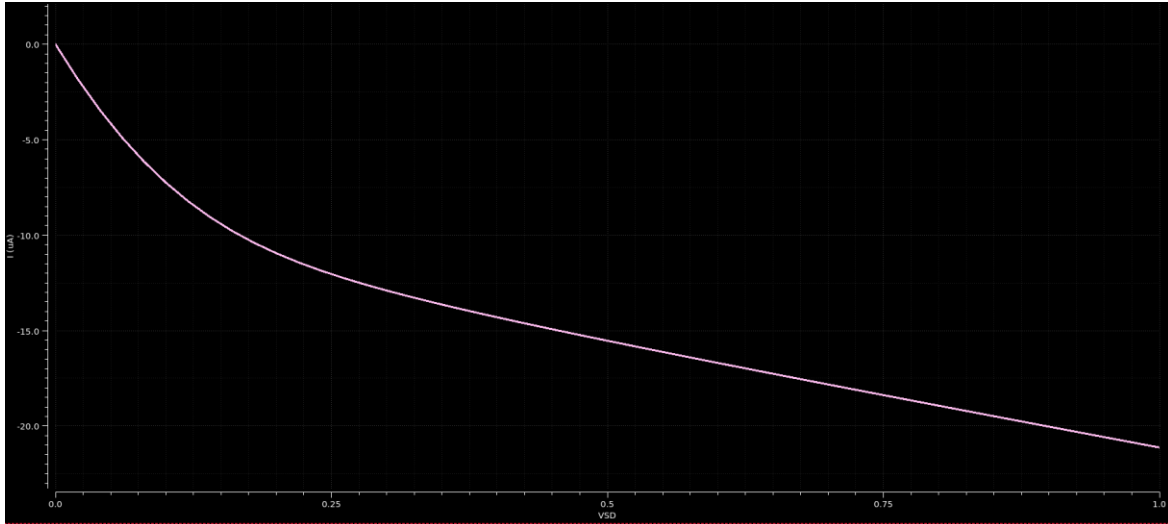


Figure 3. Output characteristic: drain current against V_{SD} at $V_{SG} = 0.8$ V.

Triode below $|V_{SD,sat}| = 209$ mV, where the channel is still continuous, then the knee into saturation as the channel pinches off near the drain.

2.3.2 Parametric sweep over V_{SG}

V_{SG} parameterized from 0 to 0.8 V in 0.2 V linear steps over the same V_{SD} sweep.

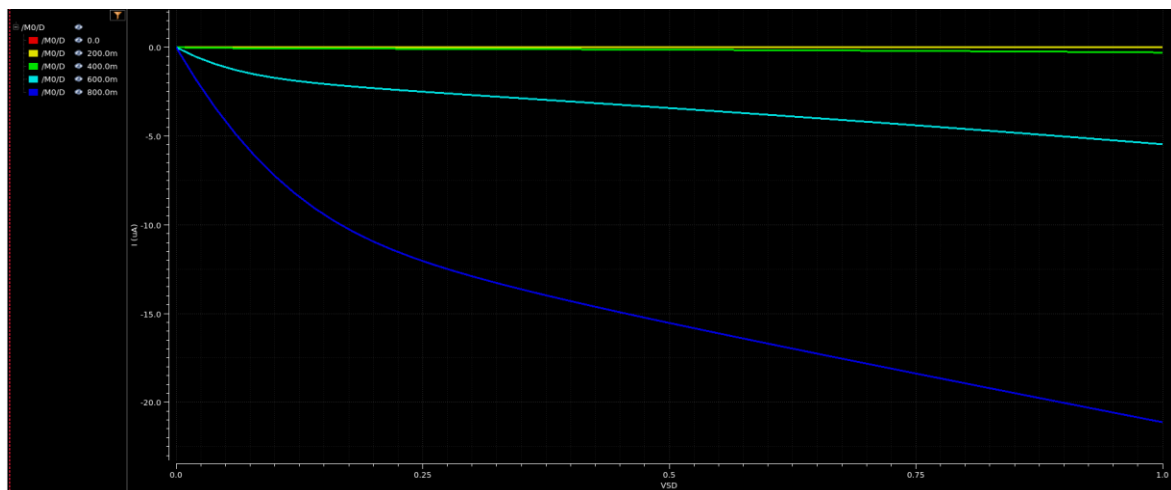


Figure 4. Output characteristics for V_{SG} stepped from 0 to 0.8 V in 0.2 V steps.

The 0, 0.2 and 0.4 V curves lie on the axis, all below $|V_{TH}| = 0.54$ V. The 0.6 V curve gives about 5.5 μ A at $V_{SD} = 1$ V and the 0.8 V curve about 22 μ A, matching Fig. 3.



2.3.3 Why the saturation region is not flat, and how to flatten it

The current should be constant past $|V_{SD,sat}|$ but climbs from about 11 μA at $V_{SD} = 0.3\text{ V}$ to 22 μA at 1 V. Two short-channel effects cause it.

Channel-length modulation. Pinch-off moves towards the source, shortening the effective channel by ΔL , so

$$I_D = I_{D,sat}(1 + \lambda V_{SD}) \quad \lambda = 1 / V_A \quad r_o = V_A / I_D$$

ΔL is set by the drain depletion region and hardly depends on L , so λ scales as $1/L$. Here $V_A = 1.28\text{ V}$, $\lambda = 0.78\text{ V}^{-1}$ and the intrinsic gain is only 6.7.

Drain-induced barrier lowering. The drain field also lowers the source barrier, cutting $|V_{TH}|$ as $|V_{SD}|$ rises. Channel-length modulation alone would give a 1.4 times rise over this range against the 2 times measured, so DIBL supplies the rest.

Remedy. Lengthen the channel and scale W to hold W/L . At $L = 240\text{ nm}$, V_A should rise about fourfold to near 5 V, cutting λ to about 0.2 V^{-1} and flattening the slope by the same factor. A cascode does the same at circuit level for the cost of one $V_{SD,sat}$ of headroom.

2.4. Activity 2.3: subthreshold conduction in the PMOS

A save_operating_points.scs file holding save M0:all was imported under Setup, Simulation Files, which makes M0:region plottable alongside I_D . The codes are 0 cut-off, 1 triode, 2 saturation, 3 subthreshold.

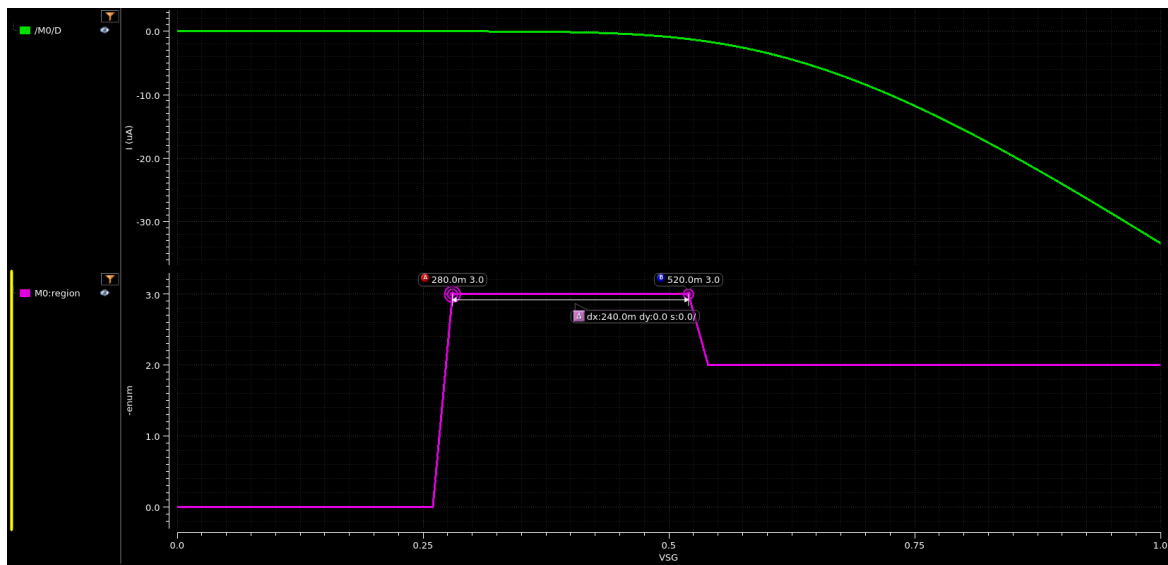


Figure 5. Drain current (upper strip) and operating-region code (lower strip) against V_{SG} . Markers A and B bracket the subthreshold window, 240 mV wide.

Region 3 runs from $V_{SG} = 0.28\text{ V}$ to 0.52 V , a 240 mV window ending at $|V_{TH}| = 537\text{ mV}$. The surface is weakly inverted there and conduction is by diffusion, so the current is exponential:



$$I_D = I_0 \exp[(V_{SG} - |V_{TH}|) / (nV_T)] \quad S = nV_T \ln 10$$

with $V_T = kT/q \approx 25.9$ mV, giving a decade of current every $S \approx 70$ to 100 mV. These currents are invisible on the linear axis of Fig. 5, which is why the region code is needed; a log plot of $|I_D|$ would give S from the slope. This is the leakage that sets standby current in a gate built from this device.

2.5. Activity 2.4: threshold voltage versus body bias and temperature

2.5.1 Body Effect

The bulk was taken off VDD and driven by a fourth source V3, so $V(\text{bulk}) = VDD + V_B$. Raising V_B from 0 to 1 V reverse-biases the source-bulk junction, and

$$|V_{TH}| = |V_{TH0}| + \gamma [(|2\phi_F| + V_{BS})^{1/2} - (|2\phi_F|)^{1/2}]$$

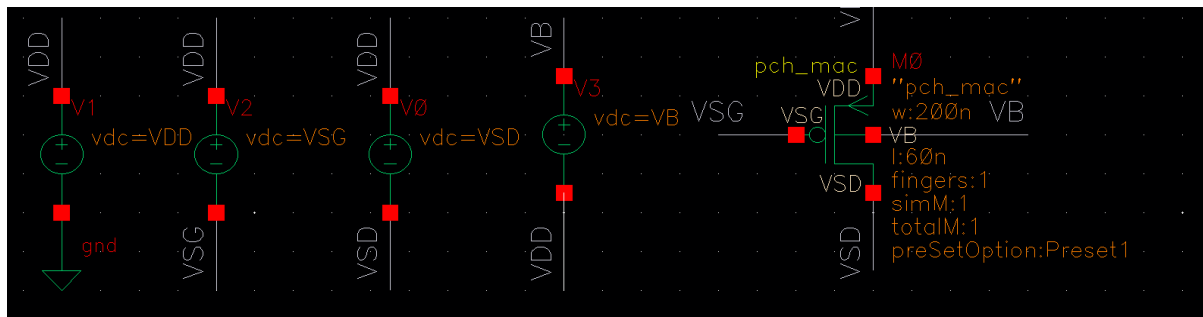


Figure 6. Testbench modified for the body-effect sweep: V3 lifts the n-well to $VDD + V_B$.

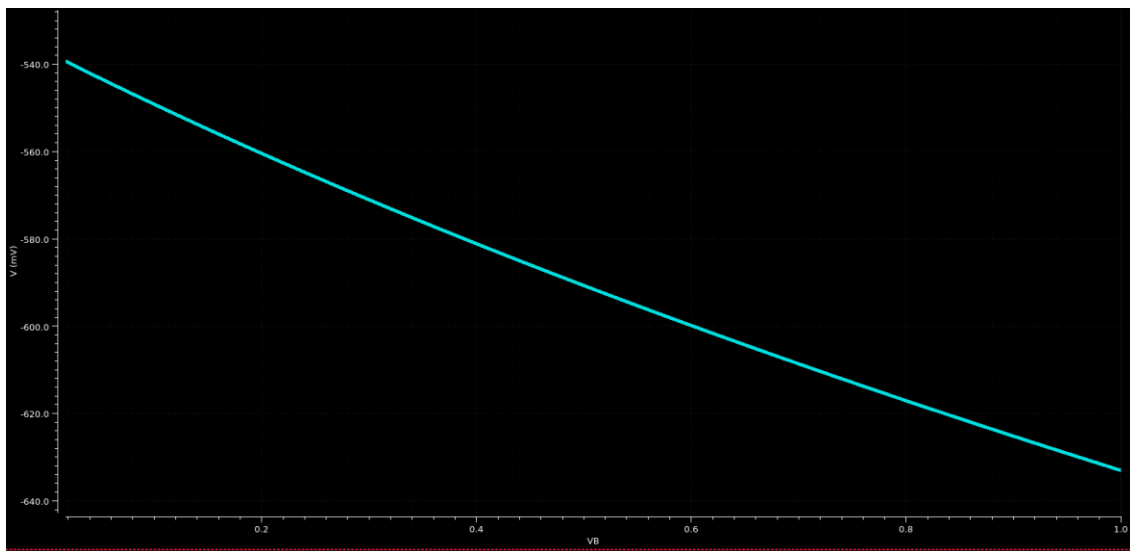


Figure 7. Threshold voltage M0:vth against body voltage V_B .

V_{TH} moves from -539 mV at $V_B = 0$ to -632 mV at $V_B = 1$ V, so $|V_{TH}|$ grows 93 mV per volt of reverse bias; with $2\phi_F \approx 0.8$ V that gives $\gamma \approx 0.21$ $V^{1/2}$. The wider depletion region needs more charge from the gate before inversion, so a stacked PMOS whose source is off the well potential loses overdrive.



2.5.2 Temperature

Same dc analysis with temperature as the sweep variable, -40 to 125 °C.



Figure 8. Threshold voltage M0:vth against temperature, -40 to 125 °C.

V_{TH} rises linearly from about -589 mV to -470 mV, roughly 0.74 mV per °C, and interpolates to -539 mV at 27 °C against the -537 mV of Table 1. ϕ_F shrinks as temperature rises, so less band bending is needed to invert the surface.

Mobility falls as roughly $T^{-1.5}$ at the same time, so the two effects cancel at one bias, the zero-temperature-coefficient point: below it the drain current has a positive temperature coefficient, above it a negative one.

3. Conclusion

The pch_mac gives $|V_{TH}| = 537$ mV at 27 °C and zero body bias, with $|V_{SD,sat}|$ already below the overdrive from velocity saturation.

Saturation is far from flat: $V_A = 1.28$ V and $g_{mro} = 6.7$, with channel-length modulation and DIBL acting together. Longer L with W scaled to match is the device fix, a cascode the circuit one.

Below threshold the device still conducts over a 240 mV window, and the threshold itself moves: up 93 mV per volt of well bias ($\gamma \approx 0.21$ V^{1/2}) and down 0.74 mV per °C of heating.